

CHETAN KULKARNI

Product & Digital Analytics | Open to Remote & California | 617-331-9347 | chetanmkulkarni@gmail.com | linkedin.com/in/chetanmkulkarni

SUMMARY

Product data scientist with 8+ years delivering action-oriented insights that drive product strategy at scale. Expert in A/B test design and execution (20+ experiments), causal inference methods (staggered DiD, propensity score matching, pre-post analysis) for consumer product decision-making, and statistical modeling. Proven track record translating complex quantitative findings into measurable outcomes — from \$665M in new deposits to self-serve analytics tools that improve cross-functional data literacy. Proficient in SQL, Python, R, Looker, and Tableau. Comfortable with ambiguous, fast-paced environments.

PROFESSIONAL EXPERIENCE

Vanguard Group, Inc

Sep 2023 - Present

Lead Product Data Analyst / Data Scientist | Dallas, TX / San Diego, CA

- Developed action-oriented insights through observational causal analysis (staggered difference-in-differences, propensity score matching) to inform product strategy for Vanguard's digital consumer journeys, serving millions of members across web and mobile.
- Architected end-to-end experimentation framework; designed and executed 20+ A/B tests with power analysis and sample size estimation, driving a 5% reduction in support call rate (~\$250K+ annual savings) and 2–3% improvement in product engagement completion rate.
- Spearheaded Cash growth initiative: designed 5 targeted experiments in coordination with cross-functional partners (Product, Marketing, Finance), generating 19,000 new accounts in a single month — ~\$665M in new deposits and an estimated \$10M+ in annualized net interest revenue.
- Designed KPI frameworks and quarterly goal-setting metrics for digital product and marketing initiatives, delivering Looker and Tableau dashboards that enabled self-serve performance monitoring for senior stakeholders.
- Mentored junior analysts on A/B test design and causal inference; served as analytics advisor to leadership, communicating statistical uncertainty to inform resource allocation decisions.

Deloitte Consulting

Jul 2021 - Sep 2023

Senior Consultant - Analytics | San Diego, CA

- Applied causal inference and A/B testing — including non-inferiority experiments and staggered difference-in-differences — to measure impact of product design changes on user engagement and conversion for client financial web platforms.
- Built self-serve analytics tools and interactive dashboards in Tableau and Looker; improved cross-functional data literacy by enabling product and design teams to independently monitor key product metrics.
- Developed KPI frameworks and measurement strategies for digital products, aligning analytics output to quarterly OKRs and business goals.
- Built ETL pipelines and data infrastructure using Apache Airflow and AWS Athena, supporting analytics at scale for client product workflows.

Genentech, Inc (via Pro-Unlimited)

Jan 2020 - Jul 2021

Informatics Data Analyst | South San Francisco, CA

- Developed a 5-layer CNN to classify image quality in clinical pathology slides and an LBP-based quality application — core invention in US Patent (US2022/0318979 A1, approved 2025).
- Built production ML deployment pipeline with multi-GPU environment and Apache Airflow automation; computed statistical quality metrics integrated with Tableau dashboard for clinical review.
- Collaborated with cross-functional ML and pathology teams to translate model outputs into clinical workflow decisions.

Deloitte Consulting

Feb 2015 - Apr 2018

Senior Consultant | Bangalore, India

- Analyzed large-scale public sector healthcare insurance data using SQL and Python; designed anomaly detection pipelines and led business case definition across cross-functional enterprise analytics engagements.

TECHNICAL SKILLS

Experimentation & Causal Inference: A/B Testing, Experiment Design, Power Analysis, Non-Inferiority Testing, Staggered DiD, Propensity Score Matching, Guardrail Metrics, Bayesian Methods

Stats & ML: Statistical Modeling, Forecasting, Regression, Time Series, Hypothesis Testing, ML, Deep Learning, Clustering, Uplift Modeling

Languages & BI: SQL (Expert), Python (Expert), R, Looker, Tableau, Data Studio

Engineering & Tools: Apache Airflow, dbt, ETL Design, AWS (SageMaker, Athena), GCP, BigQuery, KPI Frameworks, Metrics Development, Measurement Strategy, Git

Education: M.S. Data Analytics, Northeastern University | B.E. Information Science, VTU India

Patent: US2022/0318979 A1 (Approved 2025) — Image Quality Analysis for Artifact Detection | Genentech